



Quantum × AI for Healthcare Operations

Maximizing Use of Hospital Operating Theatres
through Advanced Prediction and Scheduling



Vision & Mission

- **Build the decision layer for healthcare systems**
Start with OT → expand to hospital-wide optimization
- **Expand towards other industries that require scheduling**
- **Precision medicine and drug discovery**

**We save costs, enhance efficiency, and increase wellbeing
—without sacrificing responsiveness to emergencies.**



Addressing Critical Pain Points I

Capacity is not the hospital bottleneck ...

... it's the ability to make decisions under uncertainty.

Conservative buffering is a tradeoff for safety against emergency cases and surgery overrun.



30% OT remain idle;
Only 50-60% active OT is effectively utilized



Surgical durations are stochastic



Emergency cases are unpredictable



Static Scheduling
+ Manual Coordination



Surgical resources are being wasted



Addressing Critical Pain Points II

An **impossibility** for classical algorithms to solve **at scale**.



Increase Revenue

Currently, 30% OT remain idle, utilization is 50-60% for used OTs



Increase revenue \$30,000/OT/day, \$5-20M/hospital/year



Decrease Patient Waiting Time

Currently 2-3 months wait time per patient



Reduce patient wait time by 20%



Decrease Staff Overtime

Currently 2-3 hours per day



Reduce staff overtime by 30%

Beyond Classical Systems:

Quantum-Augmented AI-Driven Multi-Objective Optimization under Uncertainty



Our Approach

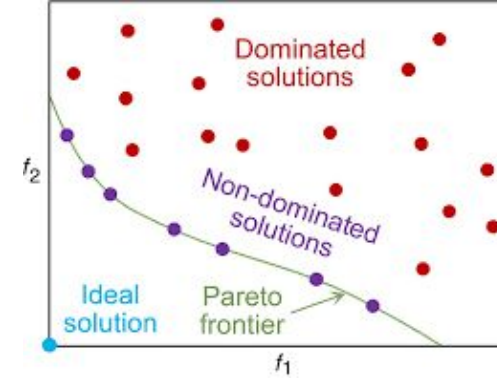
Addressing a Multi-Objective Multi-Period NP-Hard Problem under Uncertainty

1. **Forecasting**
quantum kernels $\rightarrow$ enhanced accuracy
2. **Multi-objective Optimization**
quantum computing $\rightarrow$ Pareto frontier solutions
3. **Extreme Scenario Modelling**
 $\rightarrow$ rare events, emergencies

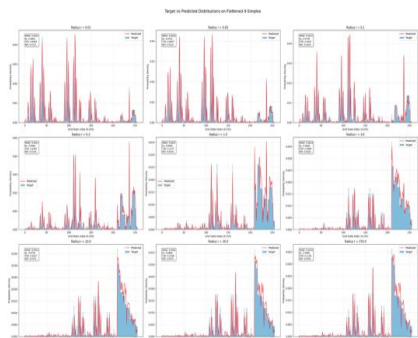
Quantum $\times$ AI Forecasting

Advanced Optimization

Hybrid Architectures
GPU + QPU



Quantum circuit configuration: $n = 8$, $N = 6$, $L = 14$, $T = 2$



At each of these 16 boundary points, we generate 10,000 samples to construct a training dataset. The experiments are implemented on the **IBM Qiskit Simulator**.

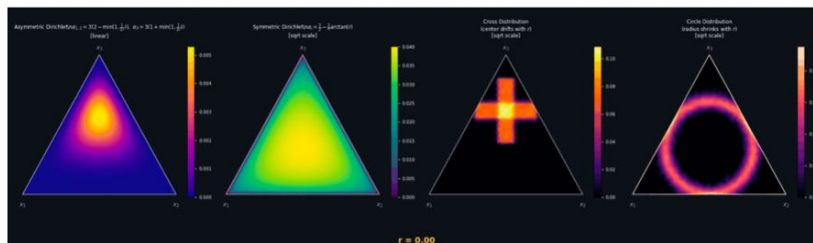
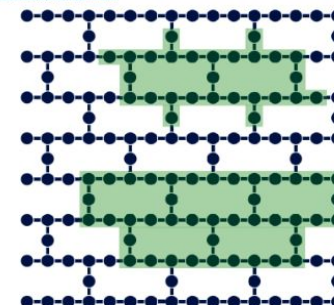


Fig. 1: Graph topologies.

From: Quantum approximate multi-objective optimization



The whole graph shows the coupling map of the 16-qubit ibm_fq device. The green-shaded areas show the 27-qubit graph of the training problem (top) and the 42-qubit graph of the considered MO-MAXCUT problem (bottom) embedded in the ibm_fq coupling map.

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Quantum approximate multi-objective optimization

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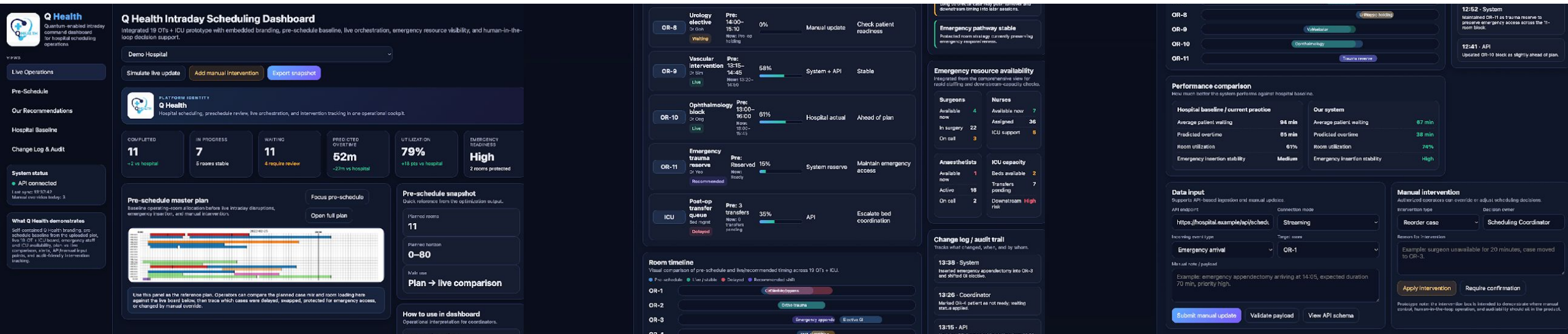
System Wide Coordination (OTs, staff & constraints)

More Accurate
Forecasting
*surgery duration &
emergencies*

Re-Optimization
real time, intra-day

Quantum-Augmented
AI Real Time Decision
Engine

Static Schedules → Real-Time Orchestration under Uncertainty

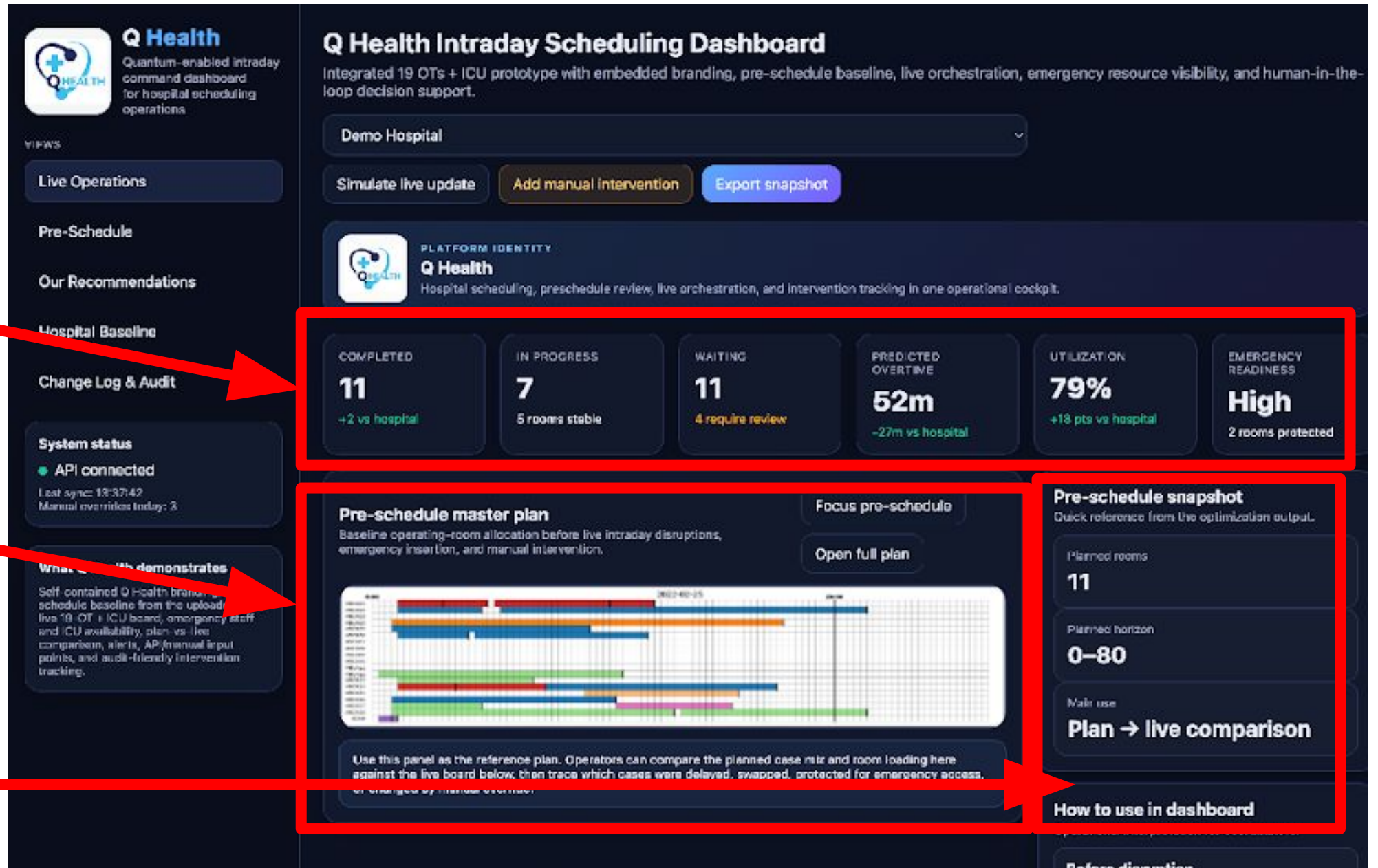




KPI to measure performance

Baseline
Operation
Theatre
Allocation

Quick
Reference from
Optimization
Output



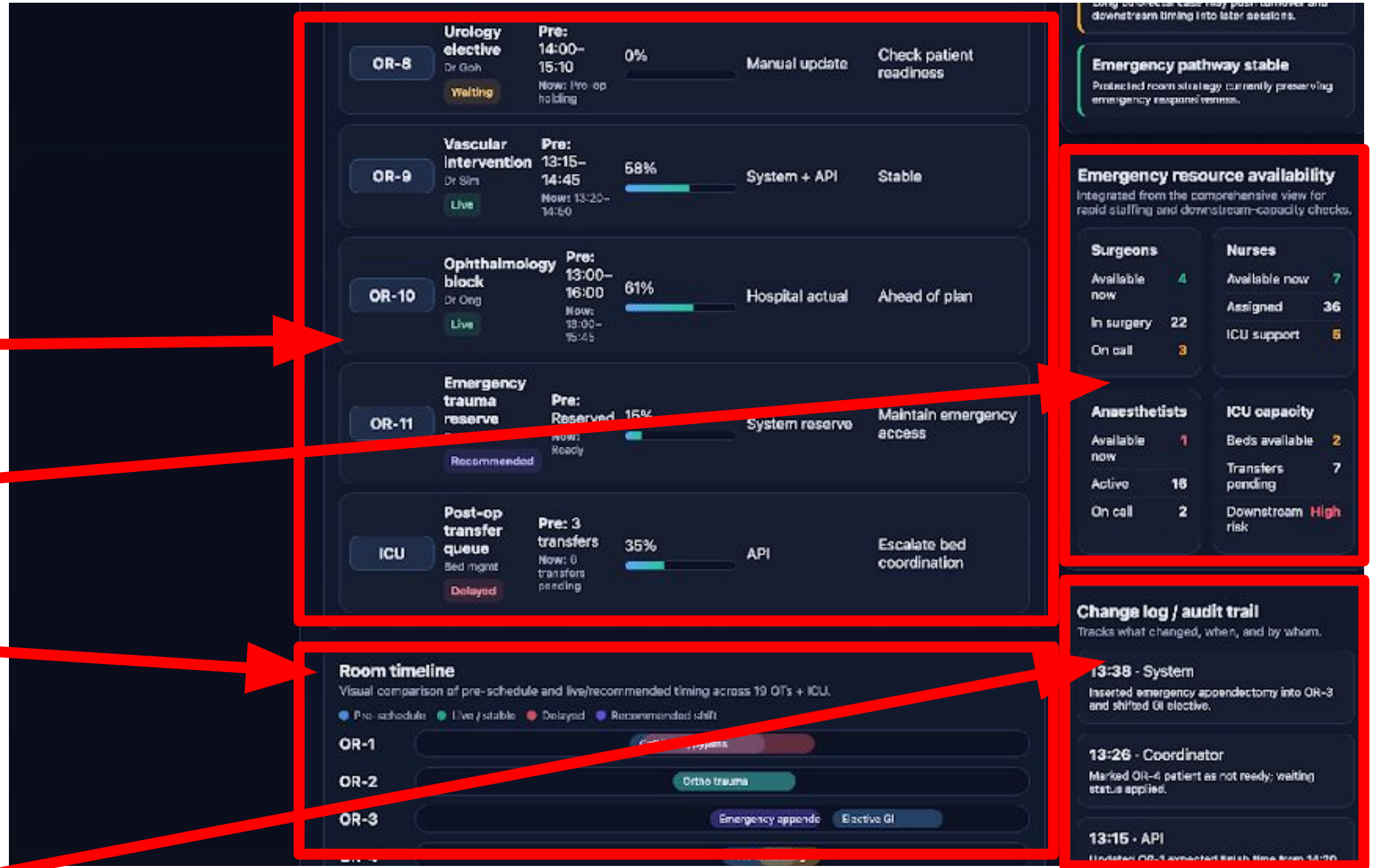


Schedule of
Each OT

Capacity
Check

Visual
Comparison of
Pre-Schedule

Tracks
Changes





How Much
Better
Q-Health
Perform

Data Input

Manual
Intervention

OR-8

Unscheduled holding

OR-9

Vascular

OR-10

Ophthalmology

OR-11

Trauma reserve

12:52 - System

Maintained OR-11 as trauma reserve to preserve emergency access across the 11-room block.

12:41 - API

Updated OR-10 block as slightly ahead of plan.

Performance comparison

How much better the system performs against hospital baseline.

Hospital baseline / current practice		Our system	
Average patient waiting	94 min	Average patient waiting	67 min
Predicted overtime	55 min	Predicted overtime	38 min
Room utilization	61%	Room utilization	74%
Emergency insertion stability	Medium	Emergency insertion stability	High

Data input

Supports API-based ingestion and manual updates.

API endpoint

https://example.com/api/v1/operations

Connection mode

Streaming

Incoming event type

Emergency arrival

Target room

OR-1

Manual note / payload

Example: emergency appendectomy arriving at 14:05, expected duration 70 min, priority high.

Submit manual update

Validate payload

View API schema

Manual intervention

Authorized operators can override or adjust scheduling decisions.

Intervention type

Reorder case

Decision owner

Scheduling Coordinator

Reason for intervention

Example: surgeon unavailable for 20 minutes, case moved to OR-3.

Apply intervention

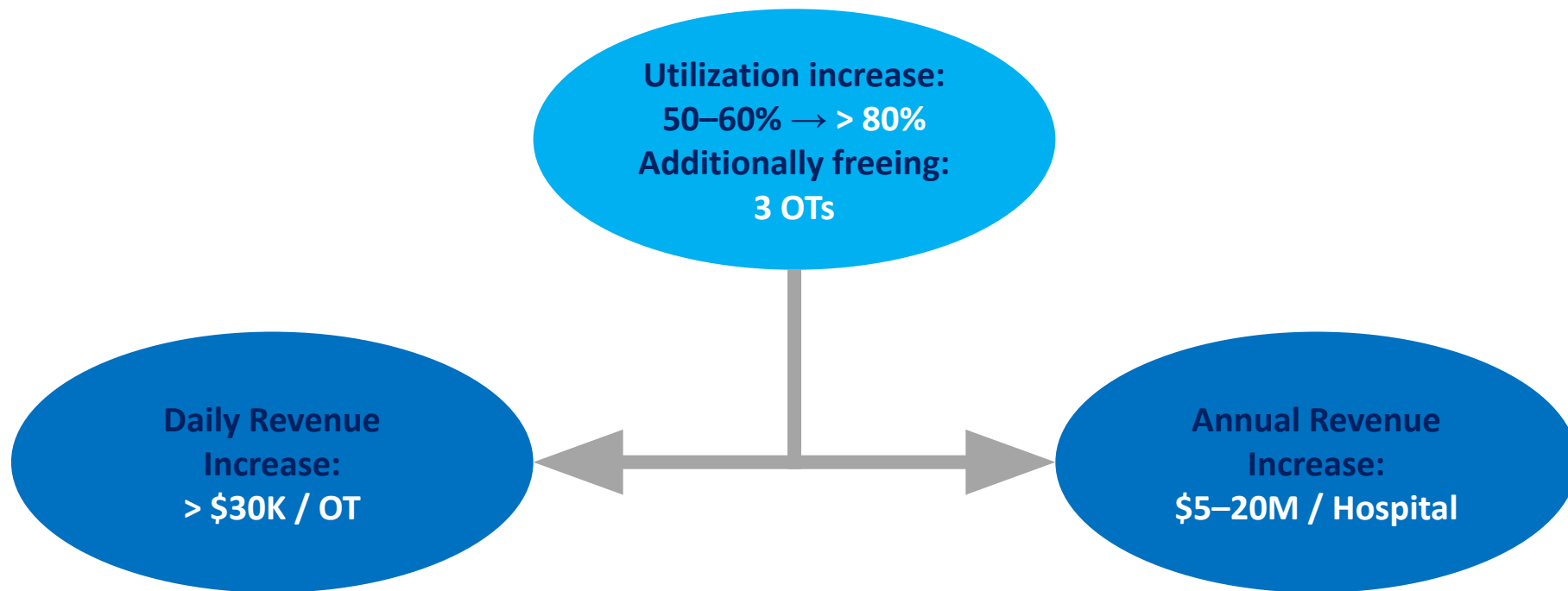
Require confirmation

Prototype note: the intervention box is intended to demonstrate where manual control, human-in-the-loop operation, and audibility should sit in the product.



Validated Impact

Real Hospital Data (19 OTs over Two Years)



Equivalent to Adding Operating Rooms Without Building New Ones



Summary of OT Outcomes & Cost-Benefit Analysis

Target Outcomes within 1 Year with 90% Confidence

- **Increase 30% Throughput**
Moving from the current 50-60% utilization baseline to >80%, effectively adding the capacity of 3 new OTs without physical construction.
- **Decrease 20% Patient Wait Time**
Addressing the current bottleneck where patients wait 6-8 months.
- **Decrease 30% Staff Overtime**
Reducing the current burden of 2-3 hours of daily overtime through better intra-day re-optimization.

Cost- Benefit Analysis

- **Performance-Based Profit Sharing**
Q-Health shares 50% of the extra profit created
- **Cost**
\$100,000 setup cost + \$15,000/OT
- **Benefit**
\$ 5-20 Million / Hospital annually



Singapore
General Hospital
SingHealth



Parkway Hospitals
SINGAPORE



Changi
General Hospital
SingHealth



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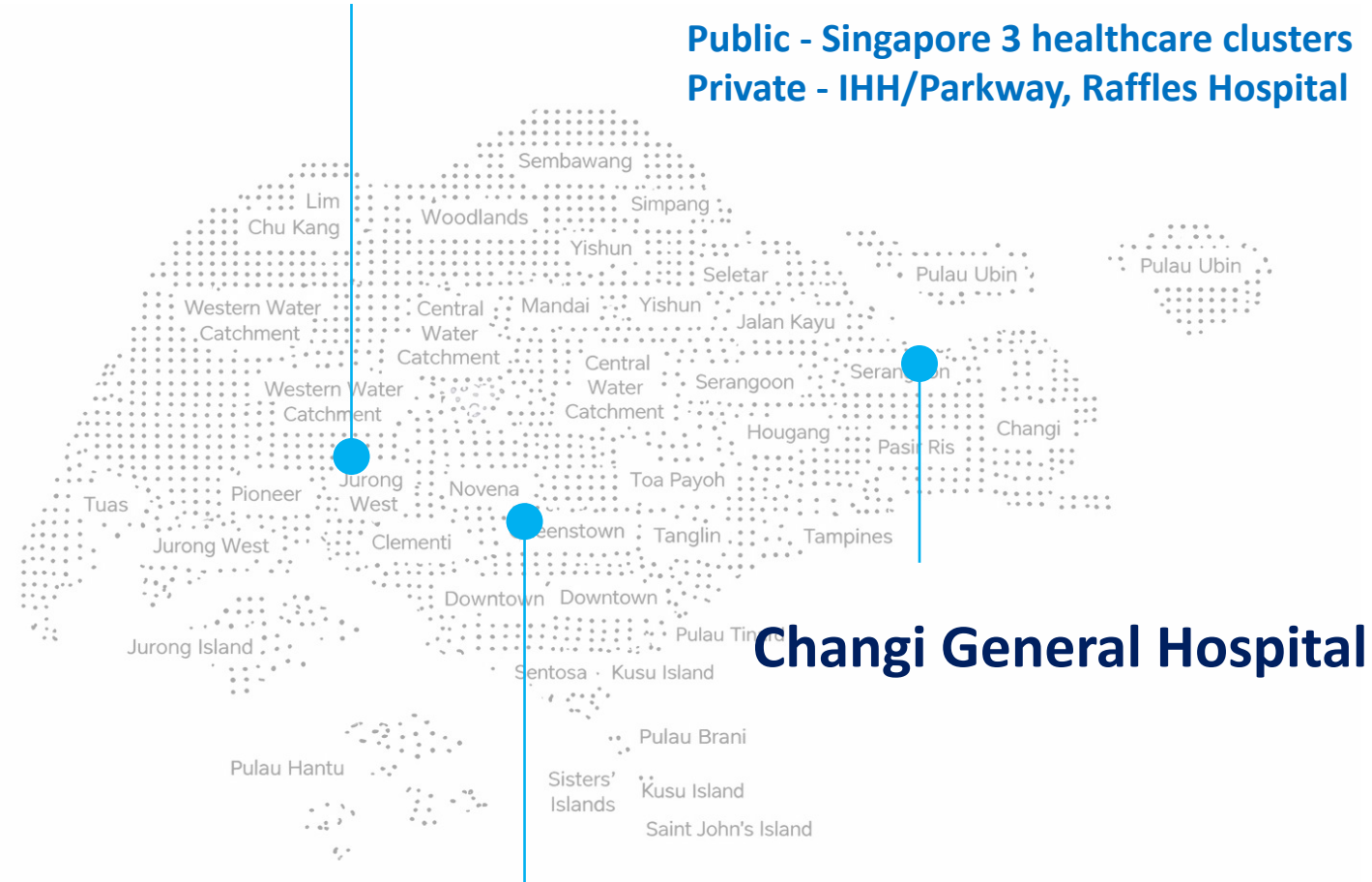
Prof. Thorsten Koch
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Target Customers

National University Hospital



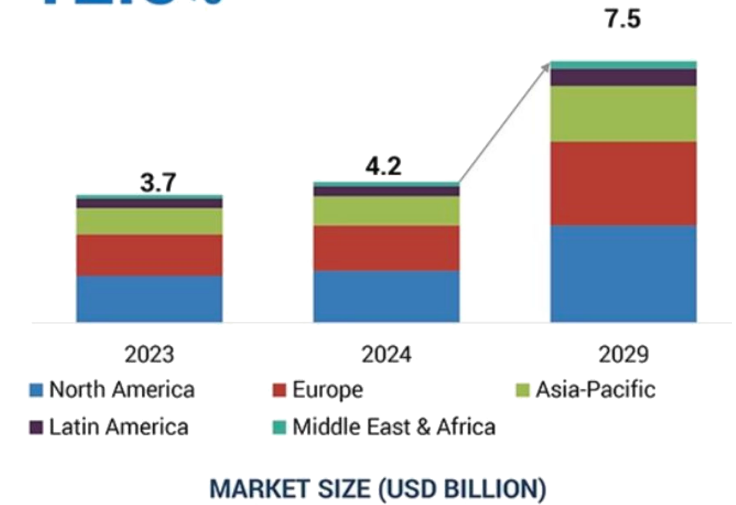
Singapore General Hospital

Provider	Market Role	Scheduling Logic	AI	Quantum Computing	Pricing Model
Q-Health	Quantum-augmented real-time prediction and scheduling engine	Multi-Objective Optimization under Multiple Constraints and Uncertainty	Predict Emergency Occurrence and Surgery Duration. Optimize OT Schedule	Quantum Readiness	\$ 90–200K + \$ 13–18K/OR/year
Epic (OpTime)	Enterprise EHR Leader	Single-Objective/Rule-Based	Predict Underutilized Block	None	\$500K-1.5M
Oracle Health	Enterprise EHR Leader	Single-Objective/Rule-Based	None	None	\$150-300K
TAGNOS	AI/IOT Real-Time OR Scheduling	Single-Objective/Rule-Based	Predict Case Length	None	\$25-100K 5-15K/OR/Year
GE Healthcare	AI Real-Time OR Scheduling	Single-Objective/Rule-Based	Advanced Scheduling, OR Capacity Planning	None	\$75-200K
Getinge (Torin)	AI Real-Time OR Scheduling	Single-Objective/Rule-Based	Improve Schedule Reliability	None	\$30-150K

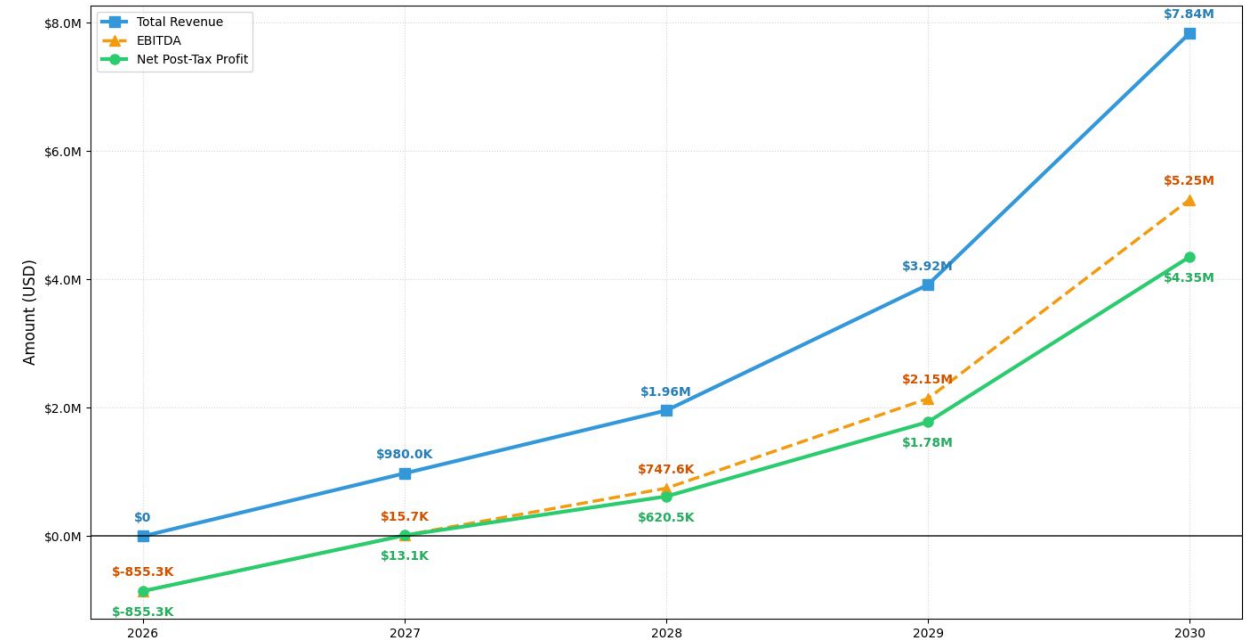


Financial Projection and KPIs

CAGR of 2024-2029
12.5%



Q-Health Financial Projection (2026-2030)



Growth Rate: **100%/year for 4 years**

Break Even Point: **2027**

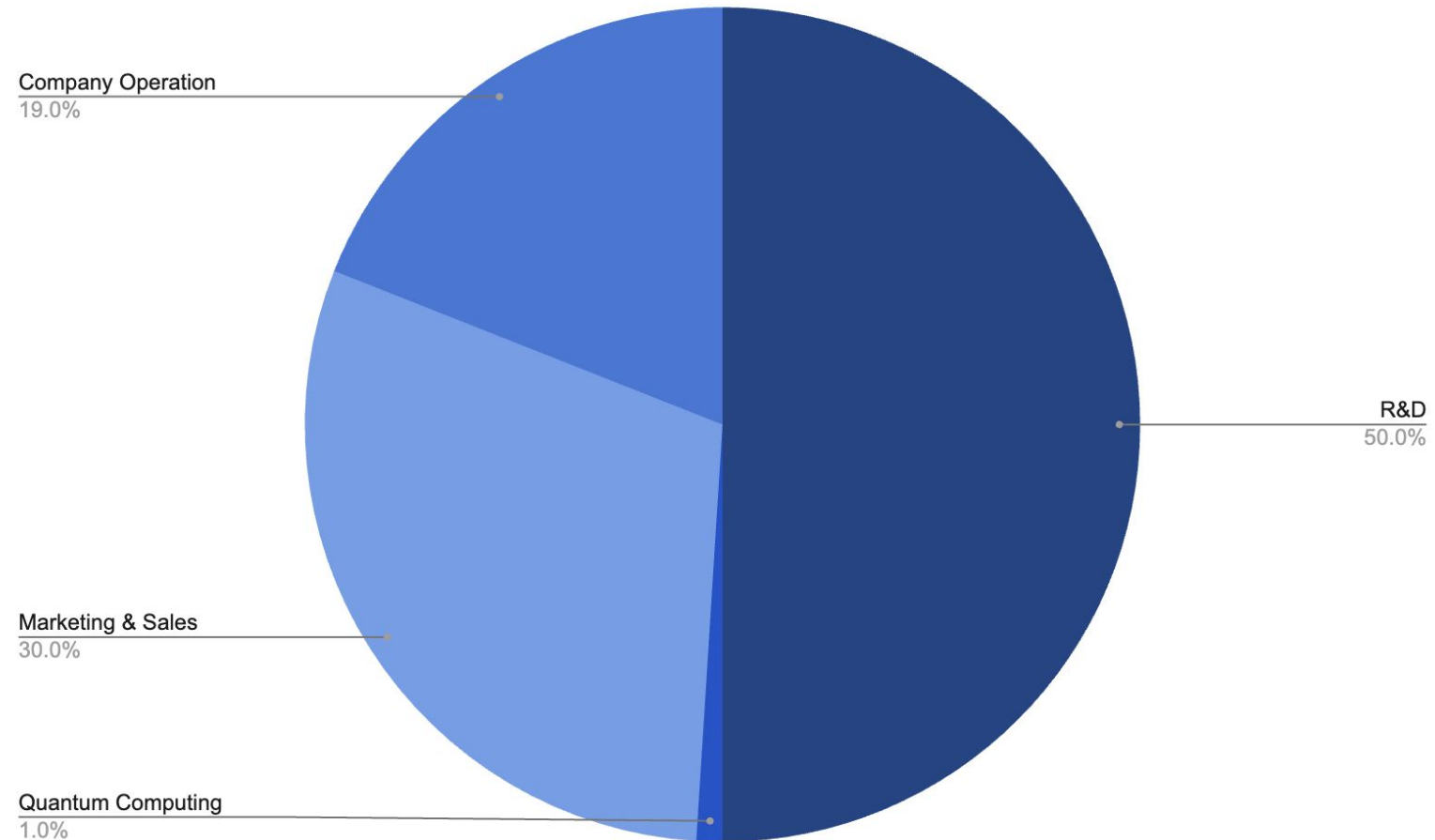
Market Size: **\$7.5B**

CAGR: **12.5%**



Asking for \$2M in Seed Round

Investment Allocation



Thank You

Any Questions?

